Sample Space and Events

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Introduction

Before any probability can be assigned, we need to know what *could* happen. The **sample space** is the set of all possible outcomes of a random experiment; **events** are subsets of the sample space to which we assign probabilities. Clean reasoning about probability begins with clear identification of both.

Prerequisites

Familiarity with basic set theory – unions, intersections, complements – is sufficient.

Theory

The **sample space** Ω is the set of all possible outcomes. Examples:

- Coin toss: $\Omega = \{H, T\}$.
- Two dice: $\Omega = \{(i, j) : i, j \in \{1, \ldots, 6\}\}$, 36 outcomes.
- Waiting time: $\Omega = [0, \infty)$.
- DNA base at position i: $\Omega = \{A, C, G, T\}$.

An **event** is a subset $A \subseteq \Omega$. We say event A occurs if the realised outcome ω is in A. Key operations:

- **Union** $A \cup B$: the event that A or B (or both) occurs.
- **Intersection** $A \cap B$: the event that both A and B occur.
- **Complement** A^c: the event that A does not occur.
- **Difference** $A \setminus B = A \cap B^c$: A but not B.

Two key identities – **De Morgan's laws** – relate complements and set operations:

$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$

Events A and B are **disjoint** (mutually exclusive) if $A \cap B = \emptyset$. They are **independent** (under probability P) if $P(A \cap B) = P(A)P(B)$. Disjointness is about sets; independence is about probabilities – the two concepts should not be confused.

Assumptions

No assumptions beyond the underlying probability space being well-defined.

R Implementation

We can enumerate finite sample spaces explicitly and manipulate events as logical vectors.

```r
# Two dice: enumerate the 36-element sample space
omega <- expand.grid(die1 = 1:6, die2 = 1:6)
head(omega, 3)

# Define events as logical vectors indexing rows of omega
A <- with(omega, die1 == 1)            # first die is 1
```

```r
B <- with(omega, die1 + die2 == 7)      # sum equals 7

# Set operations
A_union_B     <- A | B
A_inter_B     <- A & B
A_complement  <- !A
A_minus_B     <- A & !B

# Under a uniform probability on 36 outcomes:
p <- function(E) mean(E)    # proportion of outcomes in E
c(P_A         = p(A),
  P_B         = p(B),
  P_A_union_B = p(A_union_B),
  P_A_and_B   = p(A_inter_B))

# Verify inclusion-exclusion: P(A U B) = P(A) + P(B) - P(A n B)
c(lhs = p(A_union_B),
  rhs = p(A) + p(B) - p(A_inter_B))

# Verify De Morgan: (A U B)^c = A^c n B^c
all((!(A | B)) == (!A & !B))
```

Output & Results

```
  die1 die2
1    1    1
2    2    1
3    3    1

       P_A         P_B P_A_union_B   P_A_and_B
    0.1667      0.1667      0.3056     0.02778

     lhs      rhs
  0.3056   0.3056

[1] TRUE
```

All identities are satisfied. $P(A \cup B) = 11/36$ because 11 outcomes satisfy "first die is 1 OR sum is 7", and the formula recovers the same value.

Interpretation

When writing a methods section, the sample space is usually defined implicitly by the design ("patients were followed for 30 days and classified as event / no event"). Making this explicit – "the sample space is $\{\text{event}, \text{no event}\}$" – is rarely necessary except in pedagogical contexts.

Practical Tips

- For continuous sample spaces ($\Omega = \mathbb{R}$ or higher-dimensional), not every subset is an event; measurability restrictions apply. In finite or countable sample spaces, every subset is an event.
- Do not conflate *disjoint* with *independent*. Two disjoint events with positive probability are maximally dependent: if one occurs, the other cannot.
- De Morgan's laws generalise to countable unions and intersections; they are indispensable when reasoning about complements of complex events.

- Encoding finite sample spaces in R (via `expand.grid` or combinations) makes exploratory probability calculations explicit and checkable.
- Inclusion-exclusion extends to many events: $P(\bigcup A_i) = \sum P(A_i) - \sum P(A_i \cap A_j) + \ldots$.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots